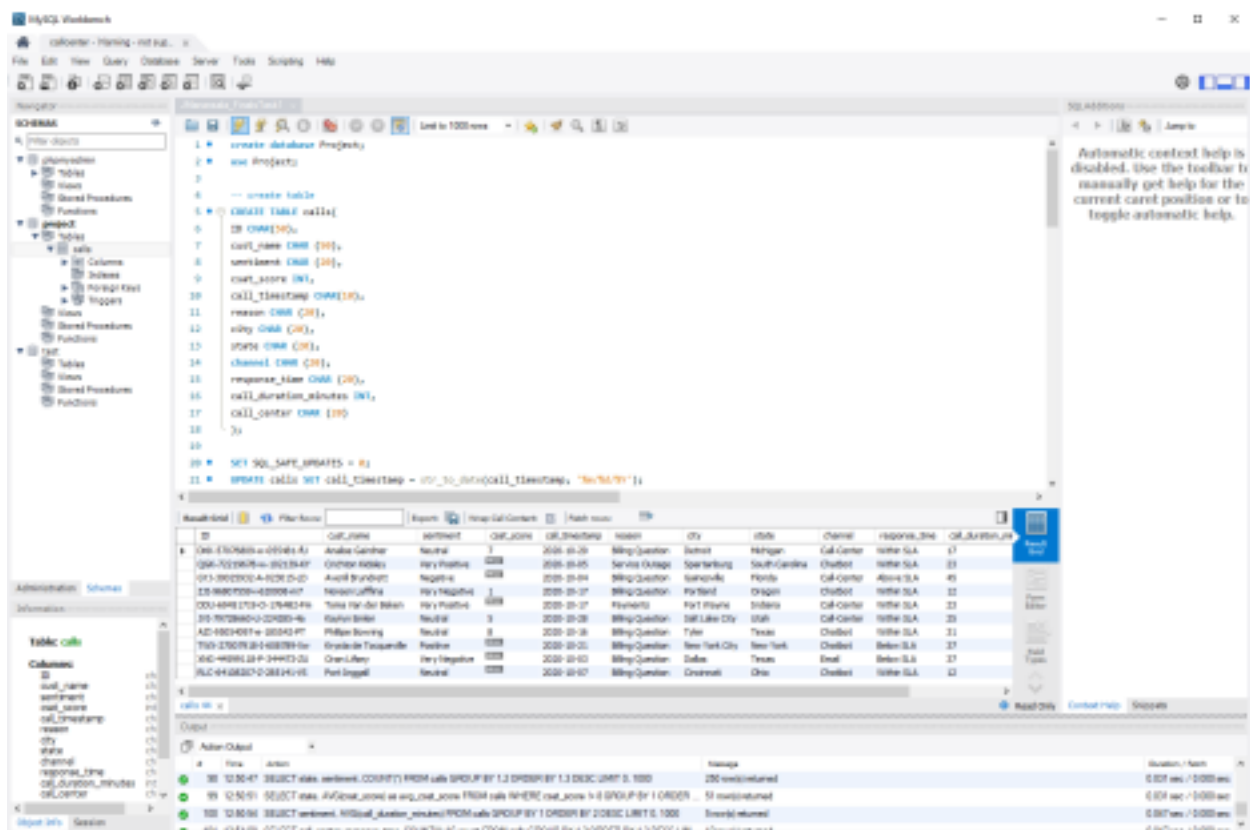


BSCS – C301

This task focuses on Data Preparation and Analysis how to clean a CSV dataset.



Monalisa_FinalsTask1

Limit to 1000 rows

```

25
26
27 -- ----- Exploring our data -----
28
29
30 -- number of columns and rows
31
32 * select count(*) as rows_num from calls;
33 * select count(*) as cols_num from information_schema.columns WHERE table_name = 'calls' ;
34
35 -- checking the distinct values of some column
36 * select distinct sentiment from calls;
37 * select distinct reason from calls;
38 * select distinct channel from calls;
39 * select distinct response_time from calls;
40 * select distinct call_center from calls;
41
42 -- the count and percentage from total of each of the distinct values
43 * SELECT sentiment, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
44 FROM calls GROUP BY 1 ORDER BY 3 DESC;
45

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

cols_num
12

Result Grid

Monalisa_FinalsTask1

Limit to 1000 rows

```

28
29
30 -- number of columns and rows
31
32 * select count(*) as rows_num from calls;
33 * select count(*) as cols_num from information_schema.columns WHERE table_name = 'calls' ;
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35 -- checking the distinct values of some column
36 * select distinct sentiment from calls;
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40 * select distinct call_center from calls;
41
42 -- the count and percentage from total of each of the distinct values
43 * SELECT sentiment, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
44 FROM calls GROUP BY 1 ORDER BY 3 DESC;
45
46 * SELECT reason, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
47 FROM calls GROUP BY 1 ORDER BY 3 DESC;
48

```

Result Grid | Filter Rows: | Export: | Wrap Cell Contents: |

call_center
Los Angeles/CA
Baltimore/MD
Denver/CO
Chicago/IL

calls 46 x

Read Only

Monarcha_FinishTask1

Limit to 1000 rows

```

52 * SELECT response_time, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
53 FROM calls GROUP BY 1 ORDER BY 3 DESC;
54
55 * SELECT call_center, count(*), ROUND((COUNT(*) / (SELECT COUNT(*) FROM calls)) * 100, 1) AS pct
56 FROM calls GROUP BY 1 ORDER BY 3 DESC;
57
58 * SELECT state, COUNT(*)
59 FROM calls GROUP BY 1 ORDER BY 2 DESC;
60
61 * SELECT dayname(call_timestamp) as Day_of_call, count(*) sum_of_calls from calls
62 GROUP BY 1 ORDER BY 2 DESC;
63
64 -- Aggregations :
65 * SELECT MIN(csaf_score) AS min_score, MAX(csaf_score) AS max_score, ROUND (AVG(csaf_score), 1) AS avg_score
66 FROM calls WHERE csaf_score != 0;
67
68 * SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls;
69
70 * SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call_du
71
72 * SELECT call_center, response_time, COUNT(*) AS count FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;

```

Result Grid

Day_of_call	sum_of_calls
Friday	5570
Thursday	5481
Wednesday	4449
Tuesday	4408
Saturday	4403
Monday	4334
Sunday	4296

Result 47

Monarcha_FinishTask1

Limit to 1000 rows

```

67
68 * SELECT MIN(call_timestamp) AS earliest_date, MAX(call_timestamp) AS most_recent FROM calls;
69
70 * SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call_du
71
72 * SELECT call_center, response_time, COUNT(*) AS count FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
73
74 * SELECT call_center, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
75
76 * SELECT channel, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
77
78 * SELECT state, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC;
79
80 * SELECT state, reason, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,2,3 DESC;
81
82 * SELECT state, sentiment, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
83
84 * SELECT state, AVG(csaf_score) as avg_csaf_score FROM calls WHERE csaf_score != 0 GROUP BY 1 ORDER BY 2 DESC;
85
86 * SELECT sentiment, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
87

```

Result Grid

sentiment	AVG(call_duration_minutes)
Negative	25.2818
Neutral	24.9398
Very Negative	24.9391
Positive	24.8620
Very Positive	24.7993

Microsoft_FinTechTask? ...

Limit to 1000 rows

```

69
70 * SELECT MIN(call_duration_minutes) AS min_call_duration, MAX(call_duration_minutes) AS max_call_duration, AVG(call_duration_minutes) AS avg_call_du
71
72 * SELECT call_center, response_time, COUNT(*) AS count FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
73
74 * SELECT call_center, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
75
76 * SELECT channel, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
77
78 * SELECT state, COUNT(*) FROM calls GROUP BY 1 ORDER BY 2 DESC;
79
80 * SELECT state, reason, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,2,3 DESC;
81
82 * SELECT state, sentiment, COUNT(*) FROM calls GROUP BY 1,2 ORDER BY 1,3 DESC;
83
84 * SELECT state, AVG(csat_score) as avg_csat_score FROM calls WHERE csat_score != 0 GROUP BY 1 ORDER BY 2 DESC;
85
86 * SELECT sentiment, AVG(call_duration_minutes) FROM calls GROUP BY 1 ORDER BY 2 DESC;
87
88 * SELECT call_timestamp, MAX(call_duration_minutes) OVER(PARTITION BY call_timestamp) AS max_call_duration
89 FROM calls GROUP BY 1 ORDER BY 2 DESC;
90

```

Result Grid | Filter Rows | Reports | Wrap Cell Contents

call_timestamp	max_call_duration
2020-10-04	46
2020-10-22	46
2020-10-26	41
2020-10-13	39
2020-10-14	37
2020-10-03	37
2020-10-30	37
2020-10-21	37
2020-10-12	35
2020-10-09	35
2020-10-25	34
2020-10-17	34

Result: 40 x 2

Result Grid | Form Editor | Field Types | Read Only